

ANTONIO LONGA — Ph.D.

✉ longaantonio@gmail.com

✉ antonio.longa@uit.no

🌐 antoniolonga.github.io

🐦 [AntonioLonga94](https://twitter.com/AntonioLonga94)

🆔 0000-0003-0337-1838 Scopus ID: 57221251335

I am a Postdoctoral Researcher at UiT The Arctic University of Norway, in the Northernmost Graph Machine Learning Group led by Filippo Maria Bianchi. My research focuses on **Graph Neural Networks**, **Network Science**, and **Complex Systems**, with particular emphasis on **Temporal Networks**, **Relational Deep Learning**, and **Explainability**. I am interested in both the **theoretical foundations** of relational and temporal learning and their **practical application** to real-world problems, including **energy systems** and **materials science** (MOF). I am an ELLIS member.

Current Position

Postdoctoral Researcher in Artificial Intelligence / Graph Machine Learning

Feb. 2026 – Present

UiT The Arctic University of Norway

Tromsø, Norway

Group: Northernmost Graph Machine Learning Group

Supervisor: Filippo Maria Bianchi

Research: Graph Machine Learning, Temporal Graphs, Relational Deep Learning, Complex Systems

Education

Ph.D. in Information & Communication Technology *magna cum laude*

Nov. 2019 – Jul. 2023

Bruno Kessler Foundation (FBK) and University of Trento

Trento, Italy

Dissertation: Understanding Social Interactions via Temporal Network Analysis

Supervisors: Andrea Passerini, Bruno Lepri

M.Sc. Computer Science, 110/110 *cum laude*

Oct. 2017 – Oct. 2019

University of Trento

Trento, Italy

Dissertation: Graph embedding in 2D

Supervisors: Fabrizio Costa, Andrea Passerini

B.Sc. Computer Science, 103/110

Sep. 2014 – Oct. 2017

University of Milano-Bicocca

Milan, Italy

Thesis: Analysis of Smali code for obfuscation detection in Android applications

Supervisors: Alberto Leporati, Claudio Ferretti

Academic Appointments & Research Experience

Assistant Professor (RTD-A) in Information & Communication Technology

Aug. 2023 – Jan. 2026

University of Trento

Trento, Italy

- Research: Relational Deep Learning, GNN Explainability, Temporal Networks.
- Teaching: Scientific Programming, Advanced Topics in Machine Learning.

Research Visitor

Jun. 2024 – Aug. 2024

Aix-Marseille University (Centre de physique théorique)

Marseille, France

- Topic: Temporal Network Generation (with Giulia Cencetti & Alain Barrat)

Research Visitor

Feb. 2024 – Mar. 2024

University of Cambridge

Cambridge, UK

- Topic: GNN regularization via explainability (with Pietro Liò)

Research Visitor

Apr. 2022 – Aug. 2022

University of Cambridge

Cambridge, UK

- Topic: GNN explainability (with Pietro Liò)

Research Trainee

Mar. 2019 – Sep. 2019

University of Exeter

Exeter, UK

- Graph embedding in low-dimensional space (with Fabrizio Costa)

Teaching Assistant

Sep. 2018 – Dec. 2018

Aalto University

Helsinki, Finland

- MSc course: Algorithmic Methods of Data Mining (with Aristides Gionis)

Machine Learning Consultant

Sep. 2021 – Present

Pulsetech (Remote)

London, UK

- GNNs for social media analytics.

Teaching & Supervision

Scientific Programming (22–25): MSc Data Science, Univ. of Trento, co-teaching, 24h

Responsibilities: delivering lab lectures, designing/marketing exams, course website with interactive materials. Topics: Python, data structures, functions, Pandas/NumPy.

Advanced Topics in ML (23–25): MSc Computer Science, Univ. of Trento (project supervision)

Responsibilities: design and supervision of tailored research projects in ML/AI. Projects aligned with students' interests, promoting engagement with state-of-the-art methods.

Machine Learning (21–22): MSc Computer Science, Univ. of Trento, co-teaching, 60h

Design/implementation of ML systems and hands-on practice (PyTorch, Scikit-learn). Responsibilities: exam design and grading, practical labs.

Computer Science (20–21): BSc Biology, Univ. of Trento, co-teaching, 36h

Introductory Python programming for biology students. Responsibilities: lab lectures, exam design and grading.

Algorithmic Methods of Data Mining (18–19): MSc Computer Science, Aalto Univ., TA, 36h

Responsibilities: grading exams and assignments.

Teaching Philosophy: Active learning (peer learning, flipped classroom), inclusivity (materials adapted to diverse backgrounds, lecture notes + tutorials), and bridging research with teaching (exposure to state-of-the-art ML, interpretability, ethical AI).

Evidence of Teaching Effectiveness: Scientific Programming (23–24) received excellent student evaluations (avg. overall satisfaction: 94/100; motivation 100/100; clarity 90/100).

Supervision: 8 MSc students (4 with accepted publications incl. ICLR, ICML workshops, LOG 2024 and Applied Network Science; 1 papers under review). Assisted in supervising 1 PhD. Projects covered GNN explainability, temporal graph generation, nonlinear sheaf diffusion, SAT solver prediction, etc. Several students pursued PhDs or positions in industry (e.g., IBM Italy).

Awards & Funding

Awards:

Covid19 PhD extension: Three paid months

NetSci2020 sponsorship: Economic support for online conference

Ph.D. scholarship: Fully funded, ranked 4th/120+

Research support UK: Seven paid months in UK

Erasmus Plus: Five paid months at Aalto University, Finland

Funding:

AIAAA – Artificial Intelligence and Advanced Analysis Applications (2021–2022)

Funded by *Legge 6 – Province of Trento*

Funding: € 350,000.

Partner: Dedagroup Business Solutions. FBK groups: DKM, MobS, Smart Communities Lab.

Role: PhD researcher (contributor to project activities).

Digitalization of the Energy System (2025–2027)

Funded by *Italian National Research Program*.

Funding: € 63,000.

Partner: Univ. of Trento & RSE.

Role: Co-PI.

Scientific Metrics

Scopus: 316 citations (h-index=10) [as of Apr 2026]

Google Scholar: 850 citations (h-index=13) [as of Apr 2026]

Publications

1. A Benchmark Dataset for Graph Regression with Homogeneous and Multi-Relational Variants

DMLR (2026)

*Hazem Peter Samoaa, Marcus Vukojevic, Morteza Haghir Chehrehghani, **Antonio Longa***

2. Actionable Real-Time Modeling of Surgical Team Dynamics via Time-Expanded Interaction Graphs

HHAI (2026)

*Vincenzo Marco De Luca, **Antonio Longa**, Giovanna Varni, Andrea Passerini*

3. Bridging Theory and Practice in Link Representation with Graph Neural Networks
NeurIPS (2025)
*Veronica Lachi, Francesco Ferrini, **Antonio Longa**, Bruno Lepri, Andrea Passerini, Manfred Jaeger*
4. Simple Path Structural Encoding for Graph Transformers
ICML (2025)
*Louis Airale, **Antonio Longa**, Mattia Rigon, Andrea Passerini, Roberto Passerone*
5. Reconsidering faithfulness in regular, self-explainable and domain invariant GNNs
ICLR (2025)
*Steve Azzolin, **Antonio Longa**, Stefano Teso, Andrea Passerini*
6. Explaining the Explainers in Graph Neural Networks: a Comparative Study
ACM Computing Survey (2025)
***Antonio Longa**, Steve Azzolin, Gabriele Santin, Giulia Cencetti, Pietro Liò, Bruno Lepri, Andrea Passerini*
7. Community Aware Temporal Network Generation
Applied Network Science (2025)
*Nicolò Alessandro Girardini, **Antonio Longa**, Gaia Trebucchi, Giulia Cencetti, Andrea Passerini, Bruno Lepri*
8. xAI-Drop: Don't Use What You Cannot Explain
Learning on Graphs (LOG 2024)
*Vincenzo Marco De Luca, **Antonio Longa**, Pietro Liò, Andrea Passerini*
9. A Simple and Expressive Graph Neural Network Based Method for Structural Link Representation
GRAM Workshop @ ICML 2024
*Veronica Lachi, Francesco Ferrini, **Antonio Longa**, Bruno Lepri, Andrea Passerini*
10. Sheaf Diffusion Goes Nonlinear: Enhancing GNNs with Adaptive Sheaf Laplacians
GRAM Workshop @ ICML 2024
*Olga Zaghen, **Antonio Longa**, Steve Azzolin, Lev Telyatnikov, Andrea Passerini, Pietro Liò*
11. Putting Context in Context: the Impact of Discussion Structure on Text Classification
EACL 2024
*Nicolò Penzo, **Antonio Longa**, Bruno Lepri, Sara Tonelli, Marco Guerini*
12. Generating fine-grained surrogate temporal networks
Communications Physics (2024)
***Antonio Longa**, Giulia Cencetti, Sune Lehmann, Andrea Passerini, Bruno Lepri*
13. A Unified Active Learning Framework for Annotating Graph Data with Application to Software Source Code Performance Prediction
Engineering Applications of Artificial Intelligence (2024)
*Peter Samoaa, Linus Aronsson, **Antonio Longa**, Philipp Leitner, Morteza Haghir Chehreghani*
14. Patterns in Temporal Networks with Higher-Order Egocentric Structures
Entropy (2023)
*Beatriz Arregui-García, **Antonio Longa**, Quintino Francesco Lotito, Sandro Meloni, Giulia Cencetti*
15. Graph Neural Networks for Temporal Graphs: State of the Art, Open Challenges, and Opportunities
Transactions on Machine Learning Research (2023)
***Antonio Longa**, Veronica Lachi, Gabriele Santin, Monica Bianchini, Bruno Lepri, Pietro Liò, Franco Scarselli, Andrea Passerini*
16. A Simple Latent Variable Model for Graph Learning and Inference
Learning on Graphs (LOG 2023)
*Manfred Jaeger, **Antonio Longa**, Steve Azzolin, Oliver Schulte, Andrea Passerini*
17. Meta-Path Learning for Multi-relational Graph Neural Networks
Learning on Graphs (LOG 2023)
*Francesco Ferrini, **Antonio Longa**, Manfred Jaeger, Andrea Passerini*
18. Global Explainability of GNNs via Logic Combination of Learned Concepts
ICLR 2023
*Steve Azzolin, **Antonio Longa**, Pietro Barbiero, Pietro Liò, Andrea Passerini*
19. An Efficient Procedure for Mining Egocentric Temporal Motifs
Data Mining and Knowledge Discovery (2022)
***Antonio Longa**, Giulia Cencetti, Bruno Lepri, Andrea Passerini*

20. Generating Synthetic Mobility Networks with Generative Adversarial Networks
EPJ Data Science (2022)
*Giovanni Mauro, **Antonio Longa**, Massimiliano Luca, Bruno Lepri, Luca Pappalardo*
21. TEP-GNN: Accurate Execution Time Prediction of Functional Tests using Graph Neural Networks
PROFES 2022
*Hazem Peter Samoa, **Antonio Longa**, Mazen Mohamad, Morteza Haghir Chehrehghani, Philipp Leitner*
22. Emotion Analysis using Multi-Layered Networks for Graphical Representation of Tweets
IEEE Access (2022)
*Anna Nguyen, **Antonio Longa**, Massimiliano Luca, Joe Kaul, Gabriel Lopez*
23. Digital Proximity Tracing on Empirical Contact Networks for Pandemic Control
Nature Communications (2021)
*Giulia Cencetti, Gabriele Santin, **Antonio Longa**, Emanuele Pigani, Alain Barrat, Ciro Cattuto, Sune Lehmann, Marcel Salathé, Bruno Lepri*

Under review:

1. Rethinking GNNs and Missing Features: Challenges, Evaluation and a Robust Solution
ICML 2026 (under review)
*Francesco Ferrini, Veronica Lachi, **Antonio Longa**, Bruno Lepri, Matono Akiyoshi, Andrea Passerini, Xin Liu, Manfred Jaeger*
2. GNNs Meet Sequence Models Along the Shortest-Path: An Expressive Method for Link Prediction
ICML 2026 (under review)
*Francesco Ferrini, Veronica Lachi, **Antonio Longa**, Bruno Lepri, Andrea Passerini*
3. Boosting Relational Deep Learning with Pretrained Tabular Models
Machine Learning (under review)
*Veronica Lachi, **Antonio Longa**, Beatrice Bevilacqua, Bruno Lepri, Andrea Passerini, Bruno Ribeiro*
4. FAIR-MOFs: A Comprehensive Database for Accelerating the Discovery and Synthesis of Metal-Organic Frameworks
Nature Materials (under review)
*Dinga Wonanke, **Antonio Longa**, Lauri Himanen, Alvin N. Ladine, Jose Marquez, Matthew A. Addicoat, Deborah Crittenden, Markus Scheidgen, Pietro Liò, Christof Woll, Thomas Heine*

Talks

Invited:

1. Modeling Time with Egocentric Temporal Structures in Networks
Aalborg University Seminar (2025)
2. The role of Egocentric Perspective in Temporal Networks
Temporal Graph Reading Group (2024)
3. Generating Temporal Networks & Journals vs. Conferences
NetPlace (2024)
4. Privacy-aware Temporal Network Generation: Methods and Applications
The MIT Club of Norway - AISD (2024)
5. Hands-on Tutorial on Graph Deep Learning
MLDS Seminar, Alan Turing Institute (2023)
6. Explaining the Explainers in Graph Neural Networks: a Comparative Study
CENTAI Seminar (2022)
7. Temporal network generation: a fast algorithm and some open problems
AIxIA Workshop on Machine Learning and Data Mining (2022)
8. Explaining the explainers in GNNs: a comparative study
GAIN Workshop: Hot topics in Graph Neural Networks (2022)
9. Neighbourhood matching creates realistic surrogate temporal networks
Cambridge Talk (2022)

Contributed:

1. Patterns in Temporal Networks with Higher-Order Egocentric Structures
NetSci (2025)

2. Constructing a Temporal Multipartite Network from News Articles
Complex Networks (2024)
3. Community aware temporal network generation
NetSciX (2024)
4. Understanding how explainers work in graph neural networks & Global Explainability of GNNs via Logic Combination of Learned Concepts
Mining and Learning with Graphs (2023)
5. Graph Neural Networks for Temporal Graphs: State of the Art, Open Challenges, and Opportunities
NetSci (2023)
6. Hands-on Tutorial on Graph Deep Learning
SIAM Conference on Computational Science and Engineering (CSE23) (2023)
7. An efficient procedure for mining egocentric temporal motifs
ECML PKDD (2022)
8. ETN-Gen: Generating Temporal Networks through Egocentric Temporal Neighbours
NetSciX (2021)
9. ETMM: Egocentric Temporal Motifs Miner
Complex Networks (2021)
10. Digital Proximity Tracing in the COVID-19 Pandemic on Empirical Contact Networks: Controlling Re-emerging Outbreaks
CCS – Complex Systems for the Most Vulnerable (2020)
11. How the Ego Perspective Shapes the Temporal Motifs in Human Face-to-Face Interactions
NetSci (2020)

Academic Service

Organizer: LOG-meetup (2023), HONS@NetSci (2025), TENET@CCS (2025), TENET@NetSci (2026)

Reviewer: NeurIPS, KDD (top reviewer), ICML, ICLR, WWW, ECML, TMLR, Neural Networks, Artificial Intelligence, ACM Computing Survey, Physics A, Machine Learning, MLG

Area Chair: LOG (2025), ICANN (2026)

Dissemination

PyTorch Geometric Tutorial: YouTube channel with 320k+ views (2019–2025)

Organizer, Pint of Science Trento (2024)

Presenter, “Networks and the Shape of Big Data”, Notte dei Ricercatori (MUSE, 2021)